

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Fifth Semester B.E. (E&E) (Credit System) Degree Examinations

January 2023

20EEE251 - FUNDAMENTALS OF PYTHON PROGRAMMING

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions by choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

Unit – I		Marks	BT*	CO*	PO*
1. a)	Explain the data types available in Python with examples.	10	L*1	1	1,2
b)	Write a program that takes the radius of a sphere (a floating-point number) as input and then outputs the sphere's diameter, circumference, surface area, and volume. Note: $V = 4/3 \pi r^3$, $A = 4 \pi r^2$, $C = 2 \pi r$, $d = 2 * r$	10	L2	1	1,2
2. a)	With algorithm or flowchart, write a python program to check whether a given number is odd or even.	10	L3	2	1,2,5,12
b)	Evaluate the following expressions. Let $x=8$, $y=2$ i) $x+y*3$ ii) $(x+y)*3$ iii) $x**y$ iv) $x\%y$ v) $x/12.0$ vi) $x//6$	6	L3	2	1,2,5,12
c)	Define for loop with an example program.	4	L2	2	1,2,5,12
3. a)	With flowchart, write a note on nested conditionals.	8	L2	2	1,2,5,12
b)	Write a note on break operator with examples.	4	L2	2	1,2,5,12
c)	Write a Python program to find the factorial of a number provided by the user.	8	L3	2	1,2,5,12
Unit – II					
4. a)	With an example, write a note on string slicing.	4	L2	3	1,2,5,12
b)	Develop a python program to convert a number from binary to decimal.	6	L3	3	1,2,5,12
c)	Define dictionary operations. Write a code segment to print all the key and value pairs in a dictionary.	10	L2	4	1,2,5,12
5. a)	Write a note on the following string methods with examples. i) <code>islower()</code> ii) <code>len()</code> iii) <code>isalpha()</code> iv) <code>capitalize()</code>	8	L2	3	1,2,5,12
b)	With an example, write a note on list slicing with example.	8	L2	3	1,2,5,12
c)	Write a python program to iterate over items in tuples using for loop.	4	L2	4	1,2,5,12
6. a)	With algorithm, write a python program to search a key element using binary search.	10	L2	3	1,2,5,12
b)	With algorithm, write a python program for implementation of Insertion Sort.	10	L2	3	1,2,5,12
Unit – III					
7. a)	With an example, write a note on creating classes in python programming.	10	L2	5	1,2,5,12
b)	With an example, write a note on Overloading operators in python.	10	L2	5	1,2,5,12
8. a)	Write a note on the concepts of encapsulation.	10	L2	5	1,2,5,12
b)	Write a note on the concepts of inheritance.	10	L2	5	1,2,5,12

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
